

Nathan Tran

Computer Science | AI/ML

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SUMMARY

Computer Science student at NC State specializing in Artificial Intelligence and Machine Learning with experience in software engineering and AI/XR research. Developed Python tools for backend testing and data analysis at Everi and research systems using Unity, computer vision, eye tracking, and multimodal sensing. Experienced in system integration, debugging, testing, prototyping, and performance optimization.

EXPERIENCE

Undergraduate Researcher

Feb 2026 – Present

Extended Horizon Lab, NC State University · Raleigh, NC

- Integrated a ZED 2i stereo camera into a projection-based XR system, preserving webcam compatibility and improving real-time performance from approximately 19 FPS to 26 FPS.
- Independently developed an eye-tracking XR research system using Meta Quest 3 and Pupil Labs Neon XR to process gaze, fixation, and attention data for proactive AI interactions.
- Developed AI-powered XR/MR systems using Unity, computer vision, eye tracking, AI agents, and multimodal sensing.
- Planned XR hardware and deployment by evaluating cameras, projectors, computing requirements, equipment compatibility, physical setup, and budget constraints.
- Contributed to UIST and CHI-related research through PulseCoach, AR/MR user studies, prototyping, experimental evaluation, research writing, and a CHI submission video.

Software Engineering Intern

Jun 2025 – Aug 2025

Everi Holdings Inc. · Las Vegas, Nevada

- Developed Python tools for transaction parsing, automated testing, and log analysis to identify backend errors and performance issues.
- Analyzed transaction data and system logs to locate bottlenecks, troubleshoot failures, and improve system reliability.
- Developed algorithm prototypes and simulations to evaluate transaction-processing behavior and performance.
- Executed backend testing and debugging to support faster and more reliable transaction processing.

SKILLS

Languages

Python · Java · C · C++ · C# · JavaScript · SQL

Frameworks & Tools

Unity · React · Spring Boot · OpenCV · Git · SQLite

AI/XR

Computer Vision · Eye Tracking · AI Agents · Multimodal Sensing

Development

REST APIs · OOP · Data Structures · Unit Testing · Linux

EDUCATION

B.S. Computer Science

Expected 2027

North Carolina State University · Raleigh, NC

Artificial Intelligence and Machine Learning Specialization

Dean's List: Fall 2023, Spring 2025, Fall 2025

College of Engineering REU, Summer 2026